

Power Saving

This page collects the practical levers a Slocum pilot can pull to **minimize energy use** — both to stretch a normal mission and to keep a glider alive as long as safely possible while it waits for recovery (after a weight drop, a leak abort, a schedule slip, or just a long pickup window). It covers saving power **while diving** and **while drifting**, and **on the surface** as well as **below it**.

Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Rev. 1, "Emergency Recovery" and "G3 Missions"), the UG2 community Slack, and the Teledyne Webb Research user forum. Several of the steps below mirror the TWR forum's "extreme energy conservation while waiting for recovery" and "standard steps for going into low power usage and drifting" posts. This is a condensed field reference — **always simulate a low-power mission in the lab before flying it**, and defer to official Teledyne documentation for your specific glider.

Where the energy goes

Roughly, a Slocum spends energy on four things:

1. **Buoyancy pump** — by far the biggest mover; every inflection costs pump work, and more drive (cc) and faster dives cost more.
2. **Pitch (battpos) motor** — repositioning the battery to set pitch.
3. **Science** — the science computer and sensors (CTD, optode, fluorometer, altimeter, etc.).
4. **Comms and "housekeeping"** — Iridium/Freewave, GPS, the flight computer's wake cycle, and time spent awake at the surface.